Exact Stack-Alphabet Costs and Nonmonotone Saturation Gaps in Guarded Two-Layer Graph Actions

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Abstract

We investigate exact realizations of guarded two-layer graph actions by one-control-state, input-consuming stack machines with top-sensitive push, pop and rewrite rules. For every root-containing upper mask over a finite two-vertex-connected graph, we determine the minimum stack-alphabet size and construct an optimal representation of height at most two. We derive the attainable population and maximum jointly realizable population at every alphabet budget; their difference exhibits a sharp nonmonotone saturation gap. At the first feasible budget the gap is exactly the vertex-cover number of an induced graph, yielding an NP-complete decision problem. A single stack cell gives an explicit linear population profile, whereas allowing a second cell activates the vertex-cover complexity. The results provide an exact resource and complexity characterization for the guarded two-layer family.

Keywords: stack-alphabet complexity; pushdown realization; labelled transition systems; protected saturation; vertex cover; rooted graph actions.
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1 Introduction

A physical realization of a labelled transition system must do more than recognize a language: it must encode every named source state and preserve the domain and result of each input action. In a pushdown implementation, such requirements impose structural constraints on the configurations available at a given stack-alphabet budget. We study those constraints for a two-layer family built from rooted, two-vertex-connected graphs.

Descriptional complexity of deterministic pushdown machines and labelled transition-system synthesis provide the general background [3, 6, 7]. Previous work on semantic normalization and exact LIFO lifting [10, 11], as well as a state–stack product lower bound for deterministic pushdown automata [8], motivate the resource-sensitive viewpoint. The present model fixes one physical control state and minimizes the stack alphabet while retaining the full labelled partial-action structure.

Main contributions

(i) An exact formula for the least stack-alphabet size of every guarded upper mask, together with a canonical realization of height two.

(ii) A complete all-budget formula for attainable population, joint saturation, and their potentially nonmonotone gap.

(iii) A vertex-cover identity and NP-completeness theorem at the first feasible alphabet budget, with a corresponding next-budget formula.

(iv) A sharp height-one versus height-two cost comparison and a complexity transition for the protected-family profile.

2 Guarded partial actions and physical devices

Let $H = (V, E)$ be a finite simple undirected two-vertex-connected graph, let $M = |V| \geq 3$, and choose a root $r \in V$. Every edge $e = \{u, v\}$ has a private input letter h_e. For a mask $S \subseteq V$ containing r, define

$$J_S = (V \times \{0\}) \cup (S \times \{1\}).$$

For $e = \{u, v\}$, the action h_e interchanges (u, b) and (v, b) when both endpoints belong to J_S and is undefined elsewhere. The vertical action z interchanges $(v, 0)$ and $(v, 1)$ exactly for $v \in S$. The root is $(r, 0)$. These actions are partial involutions, and all source states are root reachable.

A physical device consists of exactly one control state and an alphabet Γ of g stack symbols. A physical step reads one input and the current top symbol (or emptiness), then either pushes one symbol, pops the top, rewrites the top, keeps the stack unchanged, or rejects. There are no epsilon steps. A valid realization is an injective root-respecting map $\sigma : J_S \to \Gamma^*$, with $\sigma(r, 0) = \varepsilon$, which is history-independent and transports *and reflects* the domain and result of every input action. Rules depend only on the pair (input, visible top/empty); they need not be specified on unreachable configurations. Set

$$g_{\min}(H, S) = \min\{|\Gamma| : J_S \text{ admits such a realization}\}.$$

Let

$$A(S) = \{v \in S : \deg_{H[S]}(v) > 0\}, \qquad I(S) = S \setminus A(S).$$

3 An exact alphabet-size theorem

Lemma 3.1 (Stack-clique obstruction)**.** *The undirected one-step configuration graph of a one-control-state top-rewrite/push/pop stack is a block graph whose maximal cliques are $C_w = \{w\} \cup \{w\gamma : \gamma \in \Gamma\}$, for $w \in \Gamma^*$. Every finite two-vertex-connected source subgraph represented injectively by single physical steps lies inside one such clique. If it contains the empty root, it lies in C_ε.*

Proof. For any two adjacent distinct words, either one extends the other by one symbol, or both differ only in their top symbol and share the same shorter prefix. Hence every edge belongs to a prefix clique C_w. Distinct maximal prefix cliques intersect in at most one word. The resulting clique-intersection structure has no cycles of blocks, so an injective two-connected subgraph lies in a single block. The empty root appears only in C_ε. □

Theorem 3.2 (Exact alphabet cost)**.** *For every allowed rooted graph and every $r \in S \subseteq V$,*

$$\boxed{g_{\min}(H, S) = M - 1 + |A(S) \cup \{r\}| + \mathbf{1}_{\{r \in A(S),\ I(S) \neq \varnothing\}}.}$$

In particular, deciding realizability for a specified mask S and bound g takes $O(M + |E|)$ time.

Proof. The lower layer is two-vertex-connected, so Lemma 3.1 places its M distinct codes in C_ε. The root code is empty, forcing $M-1$ stack symbols. The upper root is a distinct neighbor of the empty root, so its code also lies in C_ε. If $uv \in E(H[S])$, the square on $(u,0),(v,0),(u,1),(v,1)$ attaches along its lower edge to the two-connected lower layer. The union is again two-connected. Thus every upper state indexed by $A(S)$ lies in C_ε, accounting for $|A(S) \cup \{r\}|$ additional symbols. When $r \in A(S)$ and $I(S)$ is nonempty, all the symbols already counted are used at lower states or upper states with at least one enabled horizontal action. Each upper state indexed by $I(S)$ rejects all horizontal actions; its nonempty code therefore needs a different visible top symbol, adding one symbol.

For attainment, assign the lower root to ε and every other lower state $(v,0)$ to a distinct one-symbol code c_v. Assign a fresh one-symbol code d_v to each upper state indexed by $A(S)$. If $r \notin A(S)$, assign a fresh symbol b to upper root $(r,1)$; then every other upper isolated state $(v,1)$ gets code $c_v b$. If $r \in A(S)$ but $I(S) \neq \varnothing$, introduce one further symbol b and likewise code isolated upper states by $c_v b$. The upper root is now coded by d_r. With no isolated upper states no b is necessary. All source transitions have physically adjacent codes: the lower layer uses pushes, pops, and top rewrites between ε, c_v; upper active horizontal edges use rewrites between d_u, d_v; z acts by a rewrite or a pop on the upper side; and every upper isolated state ends in b, whose horizontal instructions always reject and whose z instruction always pops. Rules on different visible tops cannot conflict, and both enabledness and targets match the source. This attains the lower bound. □

Corollary 3.3 (Depth-two optimal normal form and exact depth-one penalty)**.** *Let $g^{(h)}_{\min}(H,S)$ denote the least stack-alphabet size of a valid realization whose reachable stack height is at most h, under precisely the same single-step physical instructions and rooted source-faithfulness contract. For every $r \in S$,*

$$g^{(1)}_{\min}(H,S) = M + |S| - 1, \qquad g^{(h)}_{\min}(H,S) = g_{\min}(H,S) \quad (h \geq 2).$$

Moreover the exact penalty for insisting on stack height at most one is

$$\boxed{g^{(1)}_{\min}(H,S) - g_{\min}(H,S) = \max\{|I(S)| - 1, 0\}.}$$

Proof. The source has $M + |S|$ distinct named states, and an injective rooted realization of height at most one has only $1 + g$ possible codes, forcing $g \geq M + |S| - 1$. This is attained by coding the root by the empty word and each other source state by a different stack symbol: every legal source transition can then be implemented by one push, pop, or top rewrite, and each illegal action is rejected using its unique visible code. The constructive coding in Theorem 3.2 uses words of length at most two, so it attains $g_{\min}(H,S)$ for every $h \geq 2$. Finally, if $r \notin A(S)$ then $r \in I(S)$ and $|A(S) \cup \{r\}| = |A(S)| + 1$; if $r \in A(S)$ then the isolated states are disjoint from the root. Substitution into Theorem 3.2 proves the stated penalty in both cases. □

Remark 3.4 (Depth, rather than total stack height, can save alphabet)**.** *The precise saving comes from reusing one top marker for all but the first horizontally isolated upper states, each with a different lower prefix. Thus unbounded physical stack depth is unnecessary for the optimal alphabet count in this family: height two suffices.*

4 Protected saturation profiles

For fixed H, r, g, let $\mathcal{S}_g = \{S \subseteq V : r \in S,\ g_{\min}(H,S) \leq g\}$. If this family is nonempty, define the accessible population envelope, saturation, and family gap by

$$E_g = \bigcup_{S \in \mathcal{S}_g} J_S, \quad p_g = |E_g|, \quad \mathrm{Sat}_g = \max_{S \in \mathcal{S}_g} |J_S|, \quad \Delta_g = p_g - \mathrm{Sat}_g\,.$$

No saturation/gap is assigned when $\mathcal{S}_g = \varnothing$. These are family-profile invariants: J_S is *not* constant as S changes.

Remark 4.1 (Only the gap can be nonmonotone)**.** *Because $\mathcal{S}_g \subseteq \mathcal{S}_{g+1}$, both p_g and Sat_g are nondecreasing whenever defined. The difference $\Delta_g = p_g - \mathrm{Sat}_g$, however, need not be monotone: at the first feasible budget its population envelope can expand faster than the largest jointly realizable mask. Accordingly, the nonmonotonicity claimed in the title is a nonmonotonicity of the* gap*, never of Sat_g itself.*

Theorem 4.2 (All-budget characterization)**.** *For $g < M$, $\mathcal{S}_g = \varnothing$. For $g \geq M$,*

$$\mathrm{Sat}_g = M + \max_{S \in \mathcal{S}_g} |S|.$$

Writing $d_r = \deg_H(r)$ and $U = V \setminus N_H[r]$, the envelopes are

$$p_M = 2M - d_r, \qquad p_g = 2M \quad (g \geq M+1).$$

In particular,

$$\boxed{\Delta_M = \tau(H[U])}, \qquad \boxed{\Delta_g = M - \max_{S \in \mathcal{S}_g} |S| \quad (g \geq M+1)}.$$

For all $g \geq 2M-1$, $\Delta_g = 0$.

Proof. The empty-core mask $S = \{r\}$ costs exactly M, so the family is nonempty from that budget onward. At $g = M$, Theorem 3.2 permits precisely the independent masks $S \ni r$. These have form $\{r\} \cup T$ with T an independent subset of $H[U]$. Their union reaches upper vertices $\{r\} \cup U$ and their maximum size is $1 + \alpha(H[U])$. Thus $p_M = M + 1 + |U| = 2M - d_r$, $\mathrm{Sat}_M = M + 1 + \alpha(H[U])$, and $\Delta_M = \tau(H[U])$ by the vertex cover/independent-set identity.

At $g = M+1$, every upper vertex u becomes individually available: if u is not adjacent to r, take the independent mask $\{r, u\}$; otherwise take the adjacent two-vertex mask $\{r, u\}$, for which Theorem 3.2 gives cost $M+1$. Hence E_g contains both layers in full for every $g \geq M+1$. The saturation is the maximum source-state population by definition and the exact cost theorem. For $S = V$, $A(S) = V$, $I(S) = \varnothing$, so $g_{\min}(H, V) = 2M - 1$, yielding the last claim. □

Corollary 4.3 (A strict nonmonotone gap)**.** *For every such graph, $\Delta_{M+1} > \Delta_M$. More precisely, with $d_r = \deg_H(r)$,*

$$\Delta_{M+1} - \Delta_M = \begin{cases} d_r, & U \neq \varnothing, \\ d_r - 1, & U = \varnothing. \end{cases}$$

Thereafter Δ_g is nonincreasing for $g \geq M+1$, and first vanishes at $g = 2M-1$.

Proof. At $g = M+1$, every eligible upper mask is either independent or has exactly two adjacent vertices, one of which is r: an activated upper edge forces two upper vertices into the root clique, and any further upper vertex would require an additional root symbol or an unused horizontally silent symbol. Thus $\max|S| = \max\{1 + \alpha(H[U]), 2\}$. If U is nonempty, then $\alpha(H[U]) \geq 1$ and the maximum is $1 + \alpha(H[U])$; subtracting the expression for Δ_M gives d_r. If U is empty, the maximum is 2 and the difference is $M - 2 = d_r - 1 > 0$. For $g \geq M+1$ the envelope is constant, while the maximum feasible $|S|$ is nondecreasing, so Δ_g is nonincreasing. The full upper mask requires precisely $2M-1$ symbols by Theorem 3.2. □

Corollary 4.4 (Hardness persists at the next alphabet budget)**.** *Let $U = V \setminus N_H[r]$ and $d_r = \deg_H(r)$. At $g = M+1$, for every stack-height bound $h \geq 2$,*

$$\Delta^{(h)}_{M+1} = \begin{cases} \tau(H[U]) + d_r, & U \neq \varnothing, \\ d_r - 1, & U = \varnothing. \end{cases}$$

The decision problem whether $\Delta^{(h)}_{M+1} \le t$ is NP-complete, even when $h = 2$ is fixed. This claim concerns the variable-mask family profile, not a fixed-source assembly defect.

Proof. By Corollary 4.3, the difference between the gaps at $M+1$ and M is d_r if $U \neq \varnothing$ and $d_r - 1$ otherwise. Combine with $\Delta_M = \tau(H[U])$ and $\tau(H[\varnothing]) = 0$. For NP-hardness use the rooted graph $H(X)$ of Theorem 5.1 with a nonempty input graph X. The root has degree two and $U \cong V(X)$, so $\Delta^{(2)}_{M+1} = \tau(X) + 2$. Restriction of VERTEX COVER to nonempty graphs remains NP-complete, and the construction is polynomial. Membership in NP follows from an independent-set certificate for the corresponding bound on $\tau(X)$, or directly by the exact characterization and a vertex-cover certificate for $H[U]$. The fixed-depth-two normal form extends the result to every $h \ge 2$. □

Corollary 4.5 (Connectivity tail)**.** *Let $\kappa(H)$ be vertex connectivity. For $1 \le j \le \min\{\kappa(H), M-2\}$,*

$$\Delta_{2M-1-j} = j.$$

Proof. Delete j vertices outside the root, retaining a connected induced subgraph on $M - j \ge 2$ vertices. One may choose the deleted vertices so that the remainder is connected and contains the root: iteratively delete non-root leaves of a spanning tree, recomputing a spanning tree when necessary. The retained connected induced graph has no isolated upper vertex and hence costs $M - 1 + (M - j) = 2M - 1 - j$. This gives $\Delta \le j$. Conversely, any mask with more than $M - j$ vertices excludes fewer than $j \le \kappa(H)$ vertices. Its induced graph remains connected on at least two vertices and has cost $M - 1 + |S| > 2M - 1 - j$, a contradiction. □

5 Complexity at the first feasible budget

Theorem 5.1 (Restricted vertex-cover transfer)**.** *Given a rooted two-vertex-connected graph H and integer t, deciding whether the guarded-family gap at budget M is at most t is NP-complete.*

Proof. Theorem 4.2 gives $\Delta_M = \tau(H[V \setminus N_H[r]])$, so membership in NP follows by a standard vertex-cover certificate. For hardness, let X be an arbitrary finite simple graph. Adjoin vertices r, p, q; connect r, p, q as a triangle; connect both p and q to every vertex of X; and preserve all edges of X. The resulting graph $H(X)$ is two-vertex-connected, $N_{H(X)}[r] = \{r, p, q\}$, and $H(X)[V \setminus N[r]] \cong X$. Thus $\Delta_{|V(H(X))|} = \tau(X)$, providing a polynomial-time reduction from vertex cover. □

6 A height-induced complexity transition

For a physical stack-depth bound $h \ge 1$ and a stack-alphabet budget g, write $\mathcal{S}^{(h)}_g = \{S \ni r : g^{(h)}_{\min}(H, S) \le g\}$. Define $p^{(h)}_g$, $\mathrm{Sat}^{(h)}_g$ and $\Delta^{(h)}_g$ from the same family-envelope definitions as in Section 4. The two height regimes admit an exact comparison at *every* alphabet budget.

Theorem 6.1 (Depth-one profile and depth-two activation of vertex cover)**.** *For any rooted two-vertex-connected graph H on $M \ge 3$ vertices, $\mathcal{S}^{(1)}_g = \varnothing$ for $g < M$. At $g = M$,*

$$(p^{(1)}_M, \mathrm{Sat}^{(1)}_M, \Delta^{(1)}_M) = (M+1, M+1, 0).$$

For $g \ge M+1$,

$$\boxed{p^{(1)}_g = 2M, \qquad \mathrm{Sat}^{(1)}_g = M + \min\{M, g - M + 1\}, \qquad \Delta^{(1)}_g = \max\{0, 2M - g - 1\}.}$$

For every $h \geq 2$, these family invariants agree with the unrestricted-height profile of Theorem 4.2. In particular, at the same minimum feasible alphabet budget $g = M$*,*

$$\boxed{\Delta_M^{(1)} = 0, \qquad \Delta_M^{(h)} = \tau\big(H[V \setminus N_H[r]]\big) \quad (h \geq 2).}$$

Consequently, computing $\Delta_g^{(1)}$ is constant-time once M, g are known, whereas deciding whether $\Delta_M^{(h)} \leq t$ for variable H, r, t and any fixed $h \geq 2$ is NP-complete.

Proof. By Corollary 3.3, height one realizes precisely the masks $S \ni r$ with $M + |S| - 1 \leq g$. Since $|S| \geq 1$, no mask is realizable below M. At $g = M$ the only realizable mask is $S = \{r\}$, with $M + 1$ named states; the population envelope and saturation therefore coincide and the gap is zero. For $g \geq M + 1$, each vertex $v \neq r$ occurs in the mask $\{r, v\}$, so the population envelope contains both layers and has size $2M$. The largest feasible mask has cardinality $\min\{M, g - M + 1\}$, yielding the stated saturation and gap. The optimum depth-two construction from Corollary 3.3 proves $\mathcal{S}_g^{(h)} = \mathcal{S}_g$ for every $h \geq 2$. The first-budget identity is Theorem 4.2. The constant-time evaluation is immediate from the closed profile; NP-completeness for the depth-two (hence every $h \geq 2$) case follows from the explicit polynomial reduction in Theorem 5.1. All decision claims concern the *variable-mask family profile*, not minimization of a fixed semantic source. □

Remark 6.2 (Algorithmic interpretation)**.** *At depth one the physical alphabet has to name every semantic state; no graph optimization remains in the family profile. A second cell allows upper states with all horizontal actions disabled to share the same top marker over distinct lower prefixes; choosing which upper states are simultaneously feasible introduces maximum-independent-set optimization. Thus a second cell activates graph-dependent* family-profile *complexity, even though each fixed-mask alphabet minimum is computable in linear time. The statement is not an NP-completeness result for arbitrary one-state pushdown automata or a single fixed-source assembly defect.*

7 Related work

The physical machine reads the current input and the visible stack top; push, pop and rewrite are all available according to that pair. Accordingly, the same input letter may induce different stack operations at different source states. This is distinct from the input-partitioned transition types of visibly pushdown automata [3]. Work on accessible-stack learning [5] and bounded-height pushdown language recognition [4] provides additional automata-theoretic context.

Exact synthesis of labelled transition systems has also been studied for Petri nets and Boolean nets [6, 7]. Our target is a single-control-state stack and our measured resource is its alphabet size. Classical block-graph structure [1] supplies the geometric obstruction used in the lower bound, and classical vertex cover [9] supplies the complexity problem recovered from saturation. The author's earlier work concerns semantic normalization, LIFO lifting, and state–stack tradeoffs [10, 11, 8].

8 An illustrative four-cycle and concluding remarks

For a rooted four-cycle, the successive gaps at alphabet budgets four through seven are

$$(\Delta_4, \Delta_5, \Delta_6, \Delta_7) = (0, 2, 1, 0).$$

The attainable population grows from $p_4 = 6$ to $p_5 = 8$; the largest jointly representable population grows at a different rate. Hence although both population quantities are nondecreasing,

their difference can first increase and subsequently fall to zero. The exact alphabet cost, depth-two normal form, and vertex-cover characterization provide a unified description of this resource effect for guarded two-layer actions.

Statements and Declarations

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Data and code availability. No empirical or personal data are involved. The exact proofs and constructions are included; verification records are available in the accompanying research package.
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